

Rural Climate Data Integration

Village-Level Panel Construction | Excel Data Engineering Project

Project objective

Expand and standardize a multi-source village-year panel so the research team could analyze how extreme temperatures relate to rural population and labor mobility in China.

Pipeline

Review schema -> append historical records -> construct record keys -> match geography -> preserve 120 variables -> audit year coverage and unmatched records

Scale and output

Metric	Verified result
Original records	1,747 village-year rows
Historical records integrated	1,539 rows
Final analytical structure	3,286 rows x 120 variables
Time coverage	2009-2017
Major domains	Geography, migration, labor, income, climate, land, infrastructure

Technical contribution

Task	Implementation
Identifier design	Combined year, province code, and village code into auditable record keys
Geographic matching	Used multi-condition XLOOKUP logic to populate administrative and coordinate fields
Schema preservation	Maintained the existing 120-column research structure during row expansion
Quality control	Reconciled year counts, reviewed unmatched keys, and preserved an original-data sheet

Data governance

The source survey contains restricted village-level records, names, and coordinates. The public portfolio therefore documents aggregate coverage, field architecture, formulas, and quality-control methods without redistributing row-level data.

Skills demonstrated

Excel | XLOOKUP | Panel-data construction | Multi-source integration | Composite keys | Geographic matching | Missing-value review | Data documentation | Quality control

Role boundary

My contribution focused on data cleaning, historical-row integration, identifier reconciliation, and panel preparation. The portfolio does not claim responsibility for the project's later Stata regression stage.

Recommended next step

Move repeatable append-and-merge operations into Power Query, maintain a unique village lookup table, and add automated duplicate and unmatched-key checks before each refresh.